

CHINMAY MAHANANDA

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EDUCATION

Northeastern University

M.S. in Electrical & Computer Engineering

Seattle, WA

Sep. 2025 – May 2027

- Concentration: Hardware & Software for Machine Intelligence; Coursework: VLSI Design, Computer Vision, Advanced Computer Vision, Machine Learning & Pattern Recognition; GPA: 3.92/4.0

Dr. Ambedkar Institute of Technology

B.E. in Electronics & Communication Engineering

Bengaluru, India

Aug. 2020 – May 2024

TECHNICAL SKILLS

Languages: Verilog, Python, Embedded C, C++, MATLAB, SPICE

EDA / CAD & ML Frameworks: Cadence Virtuoso, Spectre, LTspice, OpenLane, Yosys, OpenROAD, Sky130 PDK, CST Studio Suite, KiCad, Altium, PyTorch, scikit learn, XGBoost

Verification & Dev Tools: Self-checking testbenches, golden-model co-simulation, waveform debug, Icarus Verilog, Git, Logisim, Linux internals & shell, microcontrollers

PROJECTS

RISC-V Pipelined Core — Full RTL-to-GDSII ASIC Flow | Verilog, OpenLane, Sky130

Seattle, WA

Personal Project

Summer 2026

- Designed a 4-stage pipelined RV32I core (IF→EX→MEM→WB) in Verilog with hazard forwarding, load-use stalling, and branch handling, verified bit-exact against a from-scratch golden model before synthesis.
- Carried the design through a complete open-source ASIC flow (Yosys, OpenROAD, Sky130 PDK) to signed-off GDSII at 333MHz with zero DRC/LVS violations; a stretch attempt at 500MHz was honestly documented as a failed timing closure rather than discarded.

github.com/chinmaymahananda/riscv-pipelined-core

INT8 CNN Inference Accelerator (Systolic MAC Array) | Verilog, Icarus Verilog

Seattle, WA

Personal Project

Summer 2026

- Designed a parameterizable output stationary systolic MAC array in Verilog, built bottom up from a single multiply accumulate PE with generate based scaling and signed arithmetic, verified with a self checking testbench passing 20/20 randomized matrix multiplies under Icarus Verilog.
- Extended the base array into a full INT8 CNN inference accelerator (Conv1→Conv2→FC), chaining all three layers with interlayer requantization bridges; verified bit-exact end-to-end against a Python golden model across all 20 calibration samples, matching FP32 baseline accuracy.
- Core strength in action: when the Conv2 controller produced silent numerical drift on non-square input tiles, traced the fault to an edge case in the accumulator's saturation logic via waveform-level debugging, then patched and re-verified all 20/20 cases.

github.com/chinmaymahananda/systolic-mac-array

8 Bit ALU & 4:1 Multiplexer, Digital/CMOS Logic Design | Logisim, Cadence Virtuoso

Seattle, WA

Course Project

Fall 2025

- Designed an 8-bit ALU in Logisim from a single 1-bit cell upward (parallel AND/OR/XOR/NOR, two's-complement add/subtract, SLT), then built and transient-verified a 4:1 multiplexer (hierarchical 2:1 MUX) in Cadence Virtuoso/Spectre, characterizing propagation delay across all four select states.

PCB Fabrication, Assembly & Hardware Bring Up | KiCad

Bengaluru, India

Personal Project

2023

- Owned a board end to end across schematic capture, layout, fabrication, assembly, and functional test, bringing up power and signal nets on first power on.
- Isolated a failed signal net to an improperly routed ground return via systematic continuity and probe based testing rather than reflow guesswork, then documented the corrective design change for the next revision.

Pokémon TCG Card Recognition | PyTorch, YOLO11, EfficientNet

Seattle, WA

Team Project (with Jeffrey Ma, Rayne Liu)

Spring 2026

- Engineered a YOLO11 Nano OBB detector plus EfficientNet B0 triplet loss embedding pipeline delivering sub 2 second recognition over 558 card embeddings; trained on a single RTX 3050 with augmentation to close the sim-to-real gap.

Credit Card Fraud Detection Under Extreme Class Imbalance | Python, XGBoost, SMOTE

Seattle, WA

Personal Project

Fall 2025

- Benchmarked 5 ML models on 284,807 transactions at 0.17% fraud prevalence, reaching approximately 0.99 AUC-ROC via leak-free SMOTE and recall-prioritized threshold tuning under asymmetric misclassification cost.

EXPERIENCE

AppsNDevices (Smart Class Solutions)

Remote

STEM Content Developer

2025

- Built interactive, simulation-driven STEM modules deployed to 10,000+ students across government school programs in Assam and Uttarakhand, translating circuit and physics concepts into hands-on digital simulations.

Sri Chaitanya Techno School

Bengaluru, India

Physics Faculty, IIT JEE & NEET Prep

2024

- Delivered exam prep physics instruction and lab/simulation sessions, breaking electromagnetism and circuit fundamentals into first-principles explanations; earned the Best Teacher Award (2024) for measurable gains in student performance.

PUBLICATIONS, CERTIFICATIONS & HONORS

- Publication: "Design of 6G Antenna," IJNRD 2024, Vol. 9 Iss. 6 (ISSN 2456-4184) — first author.
- Certifications & Honors: Emertxe Certified Embedded Professional (ECEP), 2025; Seattle Campus Scholarship & International Impact Award, Northeastern University (2025); Best Teacher Award (2024).